Equity Analysis of Employment Accessibility Reductions Resulting From Planned 2027 BART Service Cuts: Midterm Submission

Chloe Chon, Eliot Cummings, Rin Hinosawa, & Asha Genett

November 2, 2026

MTC Project #5: Transit Fiscal Cliff

Research Question: How will BART's planned 2027 frequency reductions, 30-minute headways on the Yellow, Blue, and Orange lines and proposed station closures, affect employment accessibility for Bay Area residents? In the final assessment, we will evaluate: Are the resulting employment accessibility losses disproportionately concentrated in socioeconomically and demographically disadvantaged communities?

To define "transit access" for walk-accessible stations, we constructed three network-distance catchment zones from homes to the BART station at 250m, 500m, 800m, and 1000m.[1] In the final analysis, network-distances to and from BART stations will be converted to approximate travel times for both walk-accessible and drive-accessible stations. To examine the changes in the number of employment opportunities within future analysis, we will use 45 minute, 60 minute, and 75 minute total transit travel times (which includes travel to BART stations from homes, total time spent on BART, and travel from BART stations to employment opportunities).

Data:

BART General Transit Feed Specification (GTFS) data: Dataset used to identify station locations and will be used in future analysis to identify train headways.

Employment 2023 Census Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) Workplace Area Characteristics (WAC): The LODES data covers all of California, so we filter for the five counties in this analysis and defined low-wage employment (CE01) as jobs which pay less than $1250 per month. We aggregate the block-level employment data to census tracts using each employment block's geographic identifier, so that the data aligns with the 2023 Census TIGER/Line census geography. Some census tracts do not have corresponding LODES employment data, and these observations are assigned to 0 to avoid miscalculations in future steps.

2023 Census TIGER/Line census tract boundaries: The dataset is filtered to include the five counties present in this analysis.

Coordinate reference system: Spatial datasets are projected to a common UTM Zone 10N (ESPG:32610) coordinate reference system to measure distances in meters.

Methods:

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[1] Durand, C. P., Tang, X., Gabriel, K. P., Sener, I. N., Oluyomi, A. O., Knell, G., Porter, A. K., Oelscher, D. M., & Kohl, H. W., 3rd (2016). The Association of Trip Distance With Walking To Reach Public Transit: Data from the California Household Travel Survey. *Journal of transport & health*, *3*(2), 154–160. https://doi.org/10.1016/j.jth.2015.08.007.

Measuring Travel Time: We create separate walk and bike graphs for each BART station area for network analysis using approximate average walking speeds at 250, 500, 800, and 1000 meters and a 15 minute driving time. We project graphs to UTM Zone 10N (EPSG:32610). For each station, we filter LODES data to block groups with population-weighted centroids falling within the created catchment area. GTFS data is used to calculate expected wait and travel time from origin station to destination station. We assume riders do not plan their trips ahead of time, but just show up to the station and wait until the next viable train arrives.
Defining Employment Access: Travel time from origin to station is ignored, all census block groups located within the catchment area (drive/walk on the origin side, walk on the destination side) are in scope. Travel time cutoff is internalized through the wait time at the station and the travel time from origin to destination BART station using GTFS data. We recurse through all census block groups in the origin station catchment area and look for destination pair census block groups, indicating where the employment opportunity is located. Jobs are deemed accessible or non-accessible if they fit within a travel time-shed of 45, 60, and 75 minutes.
Modeling Service Changes (mostly in final analysis): We will modify the GTFS data to internalize proposed service cuts in future analysis. For example, for station closures in the walking model, accessibility essentially goes to 0, we ignore other potential transit options and assume that the person simply no longer has access to the job options that BART permits.

Methodological Assumptions and Limitations
Our analysis uses several assumptions that simplify the calculation of employment accessibility under the BART service reductions. First, we define station accessibility using walking and driving catchment areas based on street networks instead of the actual straight-line distance. We are considering a 0.5 mile walking catchment, which is commonly used in policy for transit oriented development. This simplification reduces our ability to provide a more thorough analysis using a variety of distances, but would significantly increase our processing time.

Our employment accessibility calculations are binary, and valued as either reachable to a job or not. This means a 44 minute commute would count the same as a 10 minute commute. The wait times assume that riders arrive at random and wait for the next viable train. This is based off of scheduled GTFS data, which does not account for delays or missed transfers.

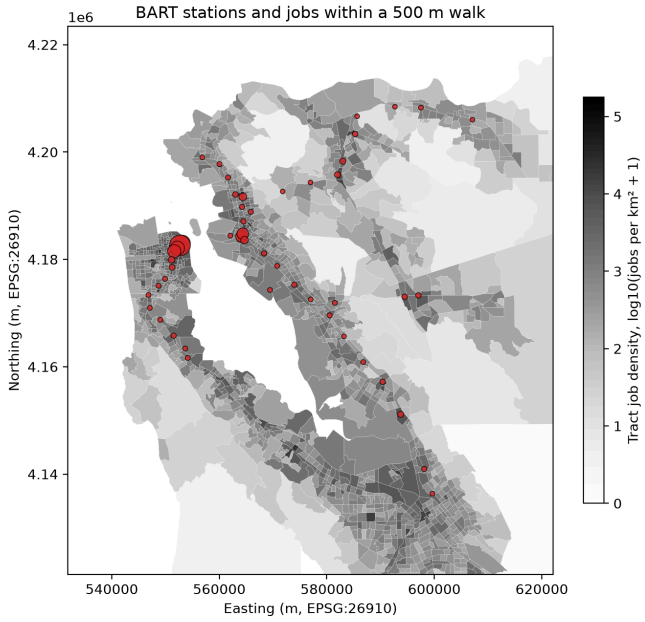
Our modeling evaluation of station closures is conservative. In our walking accessibility model, we intentionally exclude further distances residents could walk, or alternative transit options

that could change their access. In the driving model, we reassign trips to another station within a 10-minute drive and treat the trip as lost if none exists, without accounting for parking capacity.

LODES WAC measures where jobs are located, but does not account for who holds them or whether nearby residents would be qualified for them. The LODES data also excludes self-employed workers and splits the earnings in only three bands (categories). This means our definition of "low wage" is not precise, but roughly the lower ⅓ of earnings. It is also worth mentioning that OpenStreetMap walking networks ignore topography, sidewalk quality, and mobility differences.

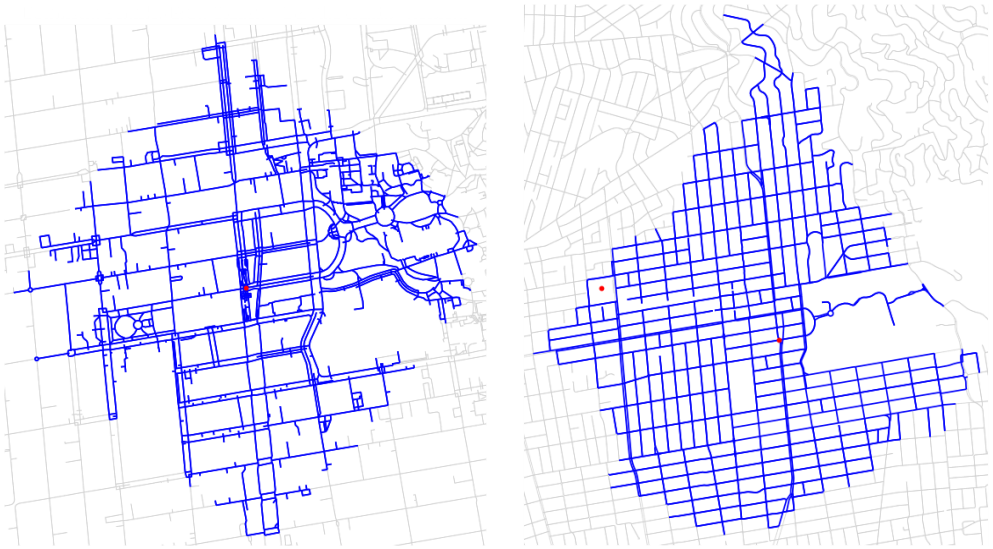
As with many limitations implicit in research, our equity findings describe overall patterns and not individual realities. Our analysis uses 2023 employment data and may not encompass the conditions present during the 2027 alternative service plan.

Early Results:

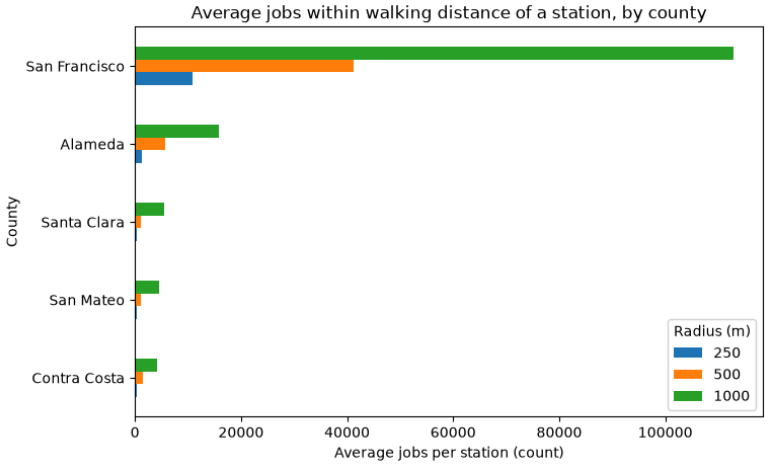


Red dot size = jobs within a 500 m straight-line walk of the station (LODES 2023). Grey shading = tract job density (log scale).

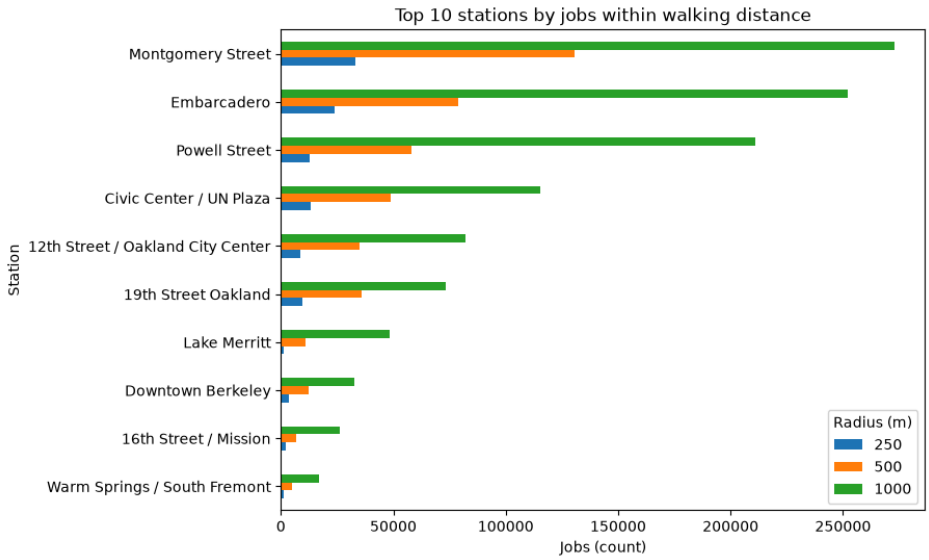
The map above shows job density from BART stations within a 500 meter walk. The next steps of our analysis will employ a similar map in order to model the changes in tacit job density after the proposed BART funding cuts (Map 1).



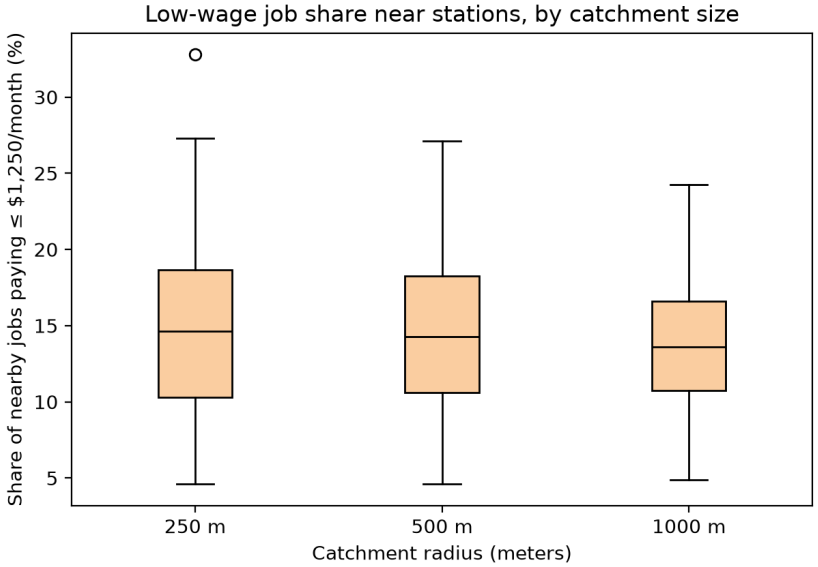
The map on the Left shows the calculated walkshed for the Downtown Berkeley BART Station station at the 800 m catchment (Map 2). The map on the Right shows the calculated driving-shed for the Downtown Berkeley BART station at the 15 minute driving shed (Map 3).



This stacked bar graph compares the total number of jobs in each county that uses the BART system, with San Francisco being the clear lead, particularly at the 1000 m catchment zone (Graph 1).



This stacked bar graph goes one step further and shows us job density for BART stations, exemplifying that stations in downtown San Francisco have the greatest density of job opportunities within their catchment (Graph 2).



One value per station. Box = middle 50% of stations, line = median. LODES 2023 earnings band CE01.

This box plot shows us the share of jobs within each catchment area from BART stations that are considered low wage, or pay less than $1250 per month (Graph 3).

Thus far, our preliminary analysis shows a substantial geographic variation in the number of employment opportunities located near BART stations. Graph 3 shows San Francisco county with over 100000 employment opportunities per station, far exceeding that of the other Bay Area counties. As shown in Graph 2, stations in downtown San Francisco have more than 5

times as many opportunities for employment located within the 1000 m catchment areas as stations in the East Bay and South Bay. Graph 3 indicates that the wage composition of employment opportunities within the catchment regions is relatively similar. Although variation across stations is present, increasing catchment size does not seem to substantially change the number of low wage employment opportunities. This information provides us with a preliminary structural understanding of the employment geography around BART stations. In the next steps of our analysis, we will employ the GTFS data to model how proposed cuts to service might affect transit users' ability to reach employment opportunities.

Plan to Finish:

Accessibility and equity analysis after changes (Nov 3 - Nov 23): All members will work to model the proposed BART station closures and decreased headways on the data already collected. This will include calculating station to station travel times, estimating employment accessibility within 45, 60, and 75 minute travel thresholds, and utilizing demographic data to examine the effects of these closures through an equity lens.

Complete visualizations and GitHub (Nov 24 - Dec 8): Finish generating at least two interactive HTML maps that show the change in employment accessibility after the proposed BART station closures and decreased headways. Complete visualizations that show employment accessibility by station, travel time-shed, and examine the equity dimension with demographic information.

Finalize deliverables (Dec 9 - Dec 15): Finish and tidy up all project deliverables, including maps, graphs, GitHub, and written analysis. Make sure that code is organized, reproducible, and documented properly.

Within our current and future analysis, challenges remain around determining reasonable thresholds for distinguishing between walk-accessible and drive-accessible stations. In future analysis, we anticipate challenges around accurately estimating expected wait and travel times using the expected value of observations on the GTFS data. We also anticipate challenges around translating walk-distance catchments into walking travel times. We anticipate our complete analysis might be computationally-intensive, and may have to consider decreasing the number of distance catchments per BART station. We would appreciate feedback regarding these future questions, as well as advice on how to ethically and properly employ demographic information during the final step of our equity analysis.